UniAdvisor AI

An AI-Powered Information Navigation System for International Students

TDK Research Proposal · Dunaújvárosi Egyetem · 2025

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Essay Submission	Approximately one week before the conference (confirm exact date with supervisor)
System Status	In active development — core architecture implemented, evaluation phase planned
Keywords	AI, RAG, NLP, university information systems, international students, LLM

⚠ Status Note for Reviewer

This proposal describes a research project currently in active development. The student needs assessment (interviews and survey) has not yet been conducted and is planned for the coming weeks. The UniAdvisor AI system is being built in parallel and will be ready for evaluation before the essay submission deadline. All methodology sections are written as planned work, not completed work, which accurately reflects the state of the project at the time of registration.

Abstract

International students at Dunaújvárosi Egyetem (UoD) face a persistent and well-documented challenge: critical university information — course registration, administrative deadlines, scholarship rules, visa procedures — exists but is buried across disjointed websites, untranslated documents, and Neptun messages that are difficult to navigate without prior Hungarian language skills or institutional knowledge. The root problem is not a lack of data; it is poor structure, limited accessibility, and invisible content.

This project proposes UniAdvisor AI: an AI-driven conversational assistant that will retrieve, contextualise, and explain university information to students in plain English or Hungarian. The system is designed around a Retrieval-Augmented Generation (RAG) pipeline over a scraped and curated knowledge base, powered by Llama 3 via the Groq API. It will cover areas including Neptun, Moodle, administrative processes, housing, visa requirements, fees, and academic calendars.

This proposal outlines the research problem, objectives, planned methodology, intended system architecture, development approach, and the expected academic and practical contribution of the project.

1. Background and Motivation

1.1 The Problem Space

Universities around the world increasingly rely on digital platforms — learning management systems, student portals, and institutional websites — to communicate critical information to their students. However, the design, language, and structure of these platforms are typically optimised for domestic students and administrative staff, not for international newcomers who arrive without fluency in the local language or familiarity with local academic culture.

At Dunaújvárosi Egyetem, informal conversations with international students have revealed recurring difficulties in the following areas:

- Understanding how Neptun — the university's course registration and grade management system — works, and interpreting automated messages sent through it.
- Locating deadline information for course registration, exam sign-ups, scholarship applications, and fee payment, which is spread across multiple pages and systems.
- Navigating the English version of the university website, which contains incomplete translations, outdated content, and an inconsistent structure.
- Understanding visa extension requirements, residence permit procedures, and health insurance obligations that apply specifically to international students.
- Identifying which university office to contact for a given problem, and what documentation to prepare before visiting.

These difficulties are not caused by a shortage of information. Official documentation, policy documents, and procedural guides exist. The core issue is one of information architecture: content is fragmented, poorly indexed, inconsistently translated, and rarely surfaced to students at the moment they need it.

1.2 Why This Problem Matters

Poor information access has documented downstream effects on international student retention, academic performance, and wellbeing (Andrade, 2006; Summers & Volet, 2008). When students cannot find clear answers about registration deadlines or visa requirements, the consequences range from missed enrolment windows to administrative penalties or visa non-compliance. A system that reliably answers these questions reduces both student stress and the administrative burden on university staff who currently handle large volumes of repetitive queries.

1.3 Why AI Is an Appropriate Solution

Large Language Models (LLMs) combined with Retrieval-Augmented Generation (RAG) offer a compelling approach to this problem. Rather than requiring students to know where to look, a RAG-based assistant accepts natural-language questions and returns precise, sourced answers drawn from an up-to-date knowledge base of institutional documents. Unlike a static FAQ or a keyword search engine, a conversational AI can handle follow-up questions, clarify misunderstandings, and adapt its language to the student's proficiency and context.

This approach has been validated in knowledge-intensive question-answering tasks (Lewis et al., 2020) and has shown strong performance in domain-specific settings where the corpus is bounded and authoritative (Gao et al., 2023). Its application to the specific context of small-to-medium European universities serving international cohorts remains underexplored, making this project a timely contribution.

1.4 Positioning Within Existing Research

AI-based student support systems have been deployed at large research universities, primarily in English-speaking countries and typically using commercial APIs with large, pre-existing knowledge bases. This project differs in three important ways: it operates at a smaller institution with constrained resources, its primary institutional language is Hungarian rather than English, and it is built entirely on publicly accessible data with no institutional API access. This positions the project as a case study in low-resource, student-driven AI deployment — a model that is directly replicable by similar institutions across Central and Eastern Europe.

2. Research Questions and Hypotheses

2.1 Primary Research Question

Can a Retrieval-Augmented Generation system, built on publicly available university data and deployed without institutional API access, materially improve the ability of international students at Dunaújvárosi Egyetem to find, understand, and act on critical institutional information?

2.2 Sub-Questions

- SQ1: What are the most frequent and high-stakes information needs of international students at UoD, and where do current information channels fail to meet them?
- SQ2: Which publicly accessible data sources provide sufficient coverage of these needs, and how should they be structured for effective retrieval?
- SQ3: What retrieval strategy — keyword-based BM25 versus dense embedding retrieval — produces more accurate and relevant responses for this domain and corpus size?
- SQ4: How do students assess the usefulness, trustworthiness, and ease of use of the resulting system compared to the existing university website?

2.3 Hypotheses

The following testable hypotheses will be evaluated during the project's evaluation phase:

- H1: A RAG-based assistant trained on scraped UoD web content will answer at least 75% of a representative benchmark question set more accurately than an unaided search of the university website.
- H2: International students will rate the conversational AI interface as significantly easier to use than the current university website for finding procedural information, as measured by the System Usability Scale (SUS).
- H3: For a corpus of this size (500–2,000 document chunks), document coverage — what is included in the knowledge base — will have a greater impact on answer quality than the choice of retrieval algorithm.

3. Objectives and Scope

3.1 Research Objectives

This project pursues two primary research objectives:

- To investigate and document the information access challenges of international students at UoD through structured primary data collection, characterising the specific deficiencies in existing information channels.
- To design, build, and evaluate an AI-based information assistant that demonstrably improves student ability to locate and act on institutional information, and to contribute a replicable methodology for similar institutions.

3.2 System Objectives

The UniAdvisor AI system will be designed to:

- Ingest and index all publicly accessible English-language content on the UoD website and downloadable institutional documents.
- Deliver accurate, source-attributed answers to student questions in both English and Hungarian via a web-based chat interface.
- Route questions automatically to the appropriate university office (Study Office, International Relations Office, Finance, IT Helpdesk, Library, Student Union, and academic departments).
- Provide university staff and administrators with a dashboard showing question patterns, student satisfaction scores, and escalation requests.
- Be deployed as a publicly accessible web application capable of serving real students during the evaluation phase.

3.3 Scope and Delimitations

The system will work exclusively with publicly accessible information — no data requiring authentication, no private student records, and no Neptun API integration. The scope covers the English-language international student journey at UoD, from pre-arrival enquiries through to graduation-related procedures. The system is not intended to replace human advisors; its role is to provide first-line information support and to route complex or sensitive cases to the appropriate staff member.

4. Planned Research Methodology

The project will proceed through four sequential phases, each building on the outputs of the previous one. All primary data collection involving human participants will take place after registration and before the essay submission deadline.

4.1 Phase 1 — Student Needs Assessment

The first phase will establish an empirical foundation for both the knowledge base and the evaluation benchmark. This will involve:

- Semi-structured interviews with approximately 10–15 international students currently enrolled at UoD, recruited through international student channels and the Student Union. Interviews will ask participants to describe specific situations in which they could not find information they needed, what they tried, and what the outcome was.

- A short structured survey (approximately 10 questions, distributed online) targeting a broader sample of 30–50 international students to quantify the frequency and severity of common information gaps across different areas (Neptun, deadlines, housing, visa, etc.).
- Thematic analysis of interview transcripts and survey responses to identify the most impactful information categories. This analysis will produce a benchmark question set of approximately 50 representative questions, which will be used to evaluate the system in Phase 4.

Ethical note: all participants will provide informed consent before taking part. No personal identifying information will be stored. Participation will be voluntary and participants may withdraw at any time.

4.2 Phase 2 — Knowledge Base Construction

The knowledge base will be built from publicly accessible sources only, with no authentication or institutional API access required:

- Automated web crawling of all English-language pages on uniduna.hu/en using a purpose-built Python crawler. The crawler will recursively discover pages, extract clean text, strip navigation and footer content, and store each page with its source URL and scrape date.
- Automated download and text extraction of publicly available PDF documents (programme handbooks, scholarship rules, visa guidance, academic calendar) using the pypdf library.
- Manual review and curation of high-priority documents to verify completeness and flag outdated content.
- Structured annotation of all content by topic category and responsible university office, enabling office-level retrieval filtering.

Text will be split into overlapping 400-word chunks with 80-word overlap, stored as JSON with metadata (source URL, office category, date scraped), and indexed using BM25 (Robertson & Zaragoza, 2009). A secondary index using sentence-transformer embeddings (all-MiniLM-L6-v2) will be built in parallel to enable a controlled comparison of retrieval strategies in the evaluation phase.

4.3 Phase 3 — System Development

The UniAdvisor AI system will be developed as a full-stack web application using an iterative approach. Development is already underway, with the core architecture in place. The remaining work before the evaluation phase includes:

- Backend (Python / FastAPI): a /chat endpoint that accepts student queries, detects the relevant office category, retrieves the top-k matching chunks via BM25, assembles a context-augmented prompt, and calls the Llama 3.3-70B model via the Groq API. A multi-key rotation system (GroqKeyRotator) handles API rate limits.
- Frontend (React / Vite): a chat interface with an explicit language toggle (English / Hungarian), an office selector, conversation history, and a thumbs-up/down feedback mechanism on each AI response.
- Persistence layer (Supabase / PostgreSQL): storing user profiles, conversation logs, feedback ratings, announcements, and escalation records.
- Admin portal: a dashboard for document upload, per-office usage analytics, satisfaction score tracking, escalation management, and an audit log of all staff actions.

- Staff portal: an interface for publishing announcements (including scheduled publishing), viewing live student activity, and sending direct messages to students.

4.4 Phase 4 — Evaluation

The system will be evaluated across three complementary dimensions:

- Answer accuracy benchmark: the 50-question benchmark set produced in Phase 1 will be answered by (a) the UniAdvisor AI system and (b) a human participant using only the current UoD website, under controlled conditions. A blinded panel of two evaluators will rate each answer for correctness, completeness, and source accuracy on a 3-point scale. This will produce a direct comparison supporting or refuting H1.
- Usability study: 8–12 international students will be asked to complete a set of information-seeking tasks using first the university website and then UniAdvisor AI (or vice versa, in counterbalanced order). Sessions will be observed using a think-aloud protocol. After each condition, participants will complete the System Usability Scale questionnaire. Results will support or refute H2.
- Operational data analysis: following a soft-launch to a pilot group of international students, production logs will be analysed for question volume, office routing distribution, thumbs-up/down ratings, and escalation rate. This will provide real-world evidence of system performance and the relative demand for different information categories.

5. System Architecture

5.1 Data Flow Overview

UniAdvisor AI follows a Retrieval-Augmented Generation pipeline extended with office routing, multi-language response locking, and a role-based administrative layer. The end-to-end data flow is:

Student query → Office detection → BM25 retrieval → Prompt assembly → LLM generation → Response + sources

5.2 Component Overview

Component	Role and Implementation
Web Crawler (crawler.py)	Recursively discovers and downloads all public English pages on uniduna.hu/en. Extracts clean body text, removes navigation and boilerplate, and saves each page with its source URL and scrape timestamp.
Document Ingester (scraper.py)	Downloads and extracts text from publicly available PDFs using pypdf. Handles programme handbooks, scholarship documents, and visa guidance.
Knowledge Base (rag.py)	Stores document chunks as JSON on disk. Builds an in-memory BM25 index at startup. Supports office-level filtering so retrieval is restricted to chunks relevant to the detected office category.
Office Router	A keyword-based classifier that maps incoming queries to one of ten office categories: Study Office, IRO, Finance, IT Helpdesk, Library, Student Union, Computer Science Dept., Engineering Dept., Economics Dept., and General.

LLM Integration	Groq API with Llama 3.3-70B as the primary model and Llama 3.1-8B as a fallback. A multi-key rotation manager (GroqKeyRotator) cycles through available API keys to handle rate limits transparently.
Backend API (main.py)	FastAPI application providing endpoints for chat, document upload, announcements, feedback, escalations, campus events, student progress tracking, and the audit log.
Student Frontend (App.jsx)	React single-page application providing the chat interface, language toggle (EN/HU), office selector, onboarding tour, progress task tracker, campus events calendar, interactive campus map, and an advisor reply inbox.
Admin Portal (AdminPortal.jsx)	Role-restricted dashboard for administrators providing KPI metrics, per-office analytics, satisfaction trend charts, escalation management, audit log, and document management.
Staff Portal (StaffPortal.jsx)	Interface for teaching and administrative staff to publish announcements, view live student question activity, browse the student directory, and send direct messages.
Database (Supabase)	Cloud-hosted PostgreSQL database storing users, chat logs, feedback ratings, announcements, escalations, campus events, progress tasks, and audit entries. Falls back to in-memory storage if unavailable.

5.3 Key Design Decisions and Justifications

- BM25 as the primary retrieval method: for a corpus of this size (estimated 500–2,000 chunks after scraping), BM25 provides competitive recall at zero inference cost per query, making the system deployable on free-tier cloud infrastructure without GPU access. Its interpretability also makes it easier to debug and improve.
- Explicit language locking: the response language (English or Hungarian) is set by the user's interface toggle and is passed as a hard constraint in the system prompt. The model is explicitly instructed not to switch languages based on query content. This prevents the common failure mode of models silently switching to Hungarian when a query contains Hungarian terms.
- Office routing before retrieval: by filtering the chunk pool to the relevant office category before running BM25, the system reduces noise in retrieved context and produces more precise answers. If office-specific retrieval returns no results, the system falls back to full-corpus retrieval.
- Graceful degradation: if the Supabase connection is unavailable, the system falls back to in-memory data stores and a set of demo credentials, ensuring the /chat endpoint remains functional during outages.

6. Expected Results and Contributions

6.1 Expected Research Contributions

- A primary-source needs analysis documenting the information access failures of international students at a mid-size Hungarian university — an institutional and demographic context that is underrepresented in the existing NLP and educational technology literature.
- An empirical comparison of BM25 versus embedding-based retrieval for a domain-specific, bilingual, institutionally-constrained RAG system, contributing to the growing literature on RAG system design for specialised corpora.

- A documented, replicable methodology for building a university AI assistant from publicly accessible data only, suitable for adoption by similar institutions across Central and Eastern Europe.

6.2 Expected Practical Contributions

- A functional, deployed web application available to all international students at UoD, providing 24/7 access to accurate, sourced answers about university procedures without requiring Hungarian language proficiency.
- An administrative layer giving university staff structured visibility into student information needs, enabling evidence-based improvement of official communications and website content.
- A scalable architecture that can be extended with new data sources — programme-specific handbooks, updated scholarship rules, or Hungarian-language content — without re-engineering the core pipeline.

6.3 Longer-Term Potential

Should the system demonstrate measurable improvement in student experience during the evaluation phase, it will provide a concrete basis for institutional adoption and integration into official university communication channels. Natural extensions include a pre-arrival information module for prospective international students, a Hungarian-language knowledge base serving domestic students, and potential integration with Neptun messaging to provide context-aware notifications. The project may also contribute directly to the university's ongoing digital transformation programme.

7. Work Plan and Timeline

The project runs from February 2025 through the TDK conference on May 13, 2025. The Gantt chart below reflects the planned schedule; phases overlap where development and data collection can proceed in parallel.

Task	Feb	Mar	Apr	May
Literature review and problem scoping	■	■		
Core system architecture and backend development	■	■	■	
Web scraping and knowledge base construction		■	■	
Frontend, admin, and staff portal development		■	■	
Student interviews and survey (Phase 1)			■	
Benchmark question set construction			■	
System integration and cloud deployment			■	■

Evaluation: accuracy benchmark and usability study				■
Data analysis and results write-up				■
TDK essay (mini-thesis) writing			■	■
Presentation preparation and rehearsal				■
TDK Conference — May 13, 2025				■

8. Resources, Budget, and Ethics

8.1 Technical Stack

- Backend language and framework: Python 3.11, FastAPI, Uvicorn
- Frontend: React 18, Vite 5, Supabase JS client
- AI and retrieval: Groq API (Llama 3.3-70B, Llama 3.1-8B), sentence-transformers (all-MiniLM-L6-v2), rank-bm25
- Database and auth: Supabase (PostgreSQL, Row Level Security, Realtime)
- Deployment: Render (free tier, backend); Vercel or Render static (frontend)
- Version control and CI: GitHub private repository

8.2 Budget

The project is designed to operate at zero direct financial cost. The Groq API free tier provides sufficient request volume for both development and the evaluation phase. The Supabase free tier accommodates up to 500 MB of database storage and 50,000 monthly active users — well above what is needed for the pilot deployment. Render and Vercel free tiers provide adequate compute and hosting for a project of this scale. No hardware purchases, software licences, or paid API credits are required.

8.3 Ethical Considerations

All web scraping will target only publicly accessible pages that do not require authentication or login. No personal student data will be collected or stored without explicit informed consent. Participants in interviews and the usability study will be briefed on the purpose of the research, their right to withdraw at any time, and how their data will be used. Interview notes and survey responses will be anonymised before analysis. Chat logs stored in Supabase will not contain personally identifiable information. The system is explicitly not designed to replace human advisors for sensitive, confidential, or legally complex cases, and will direct users to the appropriate office for such matters.

9. Planned TDK Presentation Structure

The conference presentation will be approximately 15–20 minutes, followed by a question-and-answer session. The proposed slide structure is:

#	Slide	Content
1	**Title and Introduction**	Project name, your name, supervisor, university. One-sentence problem statement.
2	**The Problem**	Evidence of information fragmentation: screenshots of unclear navigation, incomplete translations, confusing Neptun messages. Quotes from student conversations.
3	**Research Question and Hypotheses**	Primary RQ and the three hypotheses (H1, H2, H3) stated clearly.
4	**Why Retrieval-Augmented Generation?**	Plain explanation of how RAG works and why it is better suited than a static FAQ, a keyword search, or a general chatbot for this use case.
5	**Data Sources and Knowledge Base**	What was scraped, how many pages, what document types, how content is chunked and indexed.
6	**System Architecture**	Clean diagram of the data flow from student query through to response. Component table.
7	**Live System Demo**	Screen recording or live demonstration: a student asks a question, the system retrieves, responds with source attribution, and routes to an office.
8	**Admin and Staff Portals**	Dashboard screenshots: analytics panel, satisfaction scores, escalation inbox, staff announcement tool.
9	**Evaluation Methodology**	How the accuracy benchmark and usability study will be run. What the benchmark question set looks like.
10	**Results and Hypothesis Testing**	Accuracy benchmark results, SUS usability scores, pilot usage statistics. Clear statement of which hypotheses are supported.
11	**Impact and Future Potential**	Quantified improvement over baseline. Institutional adoption path. Extension to Hungarian-language content, pre-arrival module, Neptun integration.

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